# ALGERIAN MUSIC ARTISTS APPLICATION

## Part 1: XSLT Transformation

## Part 2: Flask/CLI Application


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### Administrative Footer

**ACADEMIC YEAR:** 2024-20245
**COURSE: Semi-Structured Data**
**INSTRUCTOR:** Bouniar
**SUBMISSION DATE:** [17/04/2025]

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# I.Introduction

This project develops a comprehensive digital platform to preserve and showcase Algeria's rich musical heritage. The application features an XML database of artists, transformed via XSLT for web display, complemented by a Flask-based search interface. Our solution bridges traditional music with modern technology, making cultural content accessible to global audiences while maintaining authentic representation.

# Part 1 - XSLT Transformation

## 3.1 Objective

Create an HTML document displaying all artists and their albums using:

- `ArtistAlgerian.xml` (source file)
- `ArtistAlgerian.xsl` (transformation)

## 3.2 Key XML Code

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="ArtistAlgerian.xsl"?>
<artists>
   <!-- Style Malouf -->
   <genre name="Malouf" color="#8E44AD" icon="mosque" description="Héritage
arabo-andalou de Constantine">
      <artist>
```

```
        <name>Mohamed Tahar Fergani</name>
        <city>Constantine</city>
        <site>http://www.fergani.dz</site>
        <production>Maître du Malouf</production>
        <albums>
          <album>
            <name>Nouba Zidane</name>
            <songs>
              <song>Ya Ghzali</song>
              <song>Mawwal Ana Lih</song>
            </songs>
          </album>
        </albums>
      </artist>
      <artist>
        <name>Abdelkrim Bestandji</name>
        <city>Constantine</city>
        <site>http://www.bestandji.dz</site>
        <production>Interprète classique</production>
        <albums>
          <album>
            <name>Nouba Maya</name>
            <songs>
              <song>Talaa el Badru</song>
              <song>Ya Shadi Al Alhan</song>
            </songs>
          </album>
        </albums>
      </artist>
      <artist>
        <name>Salim Halali</name>
        <city>Annaba</city>
        <site>http://www.salimhalali.dz</site>
        <production>Pionnier du Malouf</production>
        <albums>
          <album>
            <name>Trésors du Malouf</name>
            <songs>
              <song>Mahbouni Biladi</song>
              <song>Ya Rayah</song>
            </songs>
          </album>
        </albums>
      </artist>
    </genre>

    <!-- Style Raï -->
```

```
  <genre name="Raï" color="#E74C3C" icon="microphone-alt"
description="Musique populaire de l'Ouest algérien">
    <artist>
      <name>Cheb Khaled</name>
      <city>Oran</city>
      <site>https://www.chebkhaled.dz</site>
      <production>Roi du Raï</production>
      <albums>
        <album>
          <name>Kutche</name>
          <songs>
            <song>Didi</song>
            <song>Aicha</song>
          </songs>
        </album>
      </albums>
    </artist>
    <artist>
      <name>Cheb Mami</name>
      <city>Saïda</city>
      <site>https://www.chebmami.dz</site>
      <production>Voix du Raï sentimental</production>
      <albums>
        <album>
          <name>Meli Meli</name>
          <songs>
            <song>Desert Rose</song>
            <song>Parisien du Nord</song>
          </songs>
        </album>
      </albums>
    </artist>
    <artist>
      <name>Cheb Hasni</name>
      <city>Oran</city>
      <site>http://www.chebhasni.dz</site>
      <production>Roi du Raï sentimental</production>
      <albums>
        <album>
          <name>L'amour For Ever</name>
          <songs>
            <song>Beraka</song>
            <song>Galouha El Baraka</song>
          </songs>
        </album>
      </albums>
    </artist>
```

```
  </genre>

  <!-- Style Kabyle -->
  <genre name="Kabyle" color="#27AE60" icon="mountain" description="Musique berbère de Kabylie">
    <artist>
      <name>Idir</name>
      <city>Tizi Ouzou</city>
      <site>https://www.idirmusic.com</site>
      <production>Poète kabyle</production>
      <albums>
        <album>
          <name>A Vava Inouva</name>
          <songs>
            <song>A Vava Inouva</song>
            <song>Zwit Rwit</song>
          </songs>
        </album>
      </albums>
    </artist>
    <artist>
      <name>Lounès Matoub</name>
      <city>Tizi Ouzou</city>
      <site>http://www.matoub-lounes.dz</site>
      <production>Rebelle kabyle</production>
      <albums>
        <album>
          <name>Lettre Ouverte</name>
          <songs>
            <song>Regard sur l'histoire</song>
            <song>La Maison Blanche</song>
          </songs>
        </album>
      </albums>
    </artist>
    <artist>
      <name>Takfarinas</name>
      <city>Béjaïa</city>
      <site>http://www.takfarinas.dz</site>
      <production>Innovateur kabyle</production>
      <albums>
        <album>
          <name>Yal A Menfı</name>
          <songs>
            <song>Zzwaht Uzarur</song>
            <song>A Tulawin</song>
          </songs>
```

```
            </album>
        </albums>
    </artist>
  </genre>

  <!-- Style Chaâbi -->
  <genre name="Chaâbi" color="#F39C12" icon="guitar" description="Musique populaire algéroise">
    <artist>
        <name>El Hadj M'Hamed El Anka</name>
        <city>Alger</city>
        <site>http://www.elanka.dz</site>
        <production>Maître du Chaâbi</production>
        <albums>
            <album>
                <name>Cheikh El Hasnaoui</name>
                <songs>
                    <song>El Hamdoulilah</song>
                    <song>Sghir Ou Sghira</song>
                </songs>
            </album>
        </albums>
    </artist>
    <artist>
        <name>Boudjemaâ El Ankis</name>
        <city>Alger</city>
        <site>http://www.elankis.dz</site>
        <production>Rénovateur du Chaâbi</production>
        <albums>
            <album>
                <name>Nouba Chaâbi</name>
                <songs>
                    <song>Ya Mahla El Fasslaine</song>
                    <song>Moul El Kharouba</song>
                </songs>
            </album>
        </albums>
    </artist>
    <artist>
        <name>Amar Ezzahi</name>
        <city>Alger</city>
        <site>http://www.amarezzahi.dz</site>
        <production>Voix d'or du Chaâbi</production>
        <albums>
            <album>
                <name>El Hichem</name>
                <songs>
```

```
                <song>El Hichem</song>
                <song>Ya Bent Enness</song>
            </songs>
        </album>
    </albums>
  </artist>
</genre>

<!-- Style Andalouse -->
<genre name="Andalouse" color="#3498DB" icon="music" description="Musique arabo-andalouse classique">
  <artist>
    <name>Mohamed Khaznadji</name>
    <city>Alger</city>
    <site>http://www.khaznadji.dz</site>
    <production>Maître andalou</production>
    <albums>
      <album>
        <name>Nouba Gribt</name>
        <songs>
          <song>Ya Bahi El Jamal</song>
          <song>Qaduka Al Mayyas</song>
        </songs>
      </album>
    </albums>
  </artist>
  <artist>
    <name>Fadhéla Dziria</name>
    <city>Alger</city>
    <site>http://www.dziria.dz</site>
    <production>Diva andalouse</production>
    <albums>
      <album>
        <name>Nouba Dhil</name>
        <songs>
          <song>Lamma Bada Yatathanna</song>
          <song>Ya Welfi Mériem</song>
        </songs>
      </album>
    </albums>
  </artist>
  <artist>
    <name>Nouri Koufi</name>
    <city>Constantine</city>
    <site>http://www.nourikoufi.dz</site>
    <production>Interprète andalou</production>
    <albums>
```

```
            <album>
              <name>Anthologie</name>
              <songs>
                <song>Ya Warda Talaa</song>
                <song>Ya Nasr Allah</song>
              </songs>
            </album>
          </albums>
        </artist>
    </genre>

    <!-- Style Gnawa -->
    <genre name="Gnawa" color="#16A085" icon="drum" description="Musique
spirituelle afro-algérienne">
        <artist>
          <name>Maâlem Mahmoud Guinea</name>
          <city>Oran</city>
          <site>http://www.guinea.dz</site>
          <production>Maître Gnawa</production>
          <albums>
            <album>
              <name>Gnawa Diffusion</name>
              <songs>
                <song>Bambara</song>
                <song>Salat Anbi</song>
              </songs>
            </album>
          </albums>
        </artist>
        <artist>
          <name>Maâlem Abdelkader Benthami</name>
          <city>Alger</city>
          <site>http://www.benthami.dz</site>
          <production>Gnawa traditionnel</production>
          <albums>
            <album>
              <name>Lila Gnawa</name>
              <songs>
                <song>Lalla Mira</song>
                <song>Moulay Brahim</song>
              </songs>
            </album>
          </albums>
        </artist>
        <artist>
          <name>Hasna El Becharia</name>
          <city>Béchar</city>
```

```
        <site>http://www.hasnaelbecharia.dz</site>
        <production>Diva du Gnawa</production>
        <albums>
          <album>
            <name>Djazair Johara</name>
            <songs>
              <song>Bania</song>
              <song>Soudani Manayou</song>
            </songs>
          </album>
        </albums>
      </artist>
    </genre>
</artists>
```

## 3.3 Key XSLT Code

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
version="1.0">
   <xsl:output method="html" encoding="UTF-8" indent="yes"/>

   <xsl:template match="/">
      <html lang="fr">
        <head>
          <meta charset="UTF-8"/>
          <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
          <title>Patrimoine Musical Algérien</title>
          <link
href="https://fonts.googleapis.com/css2?family=Playfair+Display:wght@400;700&
amp;family=Poppins:wght@300;400;600&amp;family=Amiri:wght@400;700"
rel="stylesheet"/>
          <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0/css/all.min.css"
/>
          <style>
            :root {
              --gold: #D4AF37;
              --dark-blue: #0A2463;
              --light-blue: #3E92CC;
              --white: #FFF;
```

```
    --light-gray: #F5F5F5;
    --dark: #1E1E24;
}

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Poppins', sans-serif;
    background-color: var(--light-gray);
    color: var(--dark);
    line-height: 1.7;
}

.container {
    max-width: 1400px;
    margin: 0 auto;
    padding: 2rem;
}

header {
    background: linear-gradient(135deg, var(--dark-blue) 0%, #1E3A8A
100%);
    color: var(--white);
    text-align: center;
    padding: 4rem 0;
    margin-bottom: 3rem;
    position: relative;
    overflow: hidden;
    box-shadow: 0 5px 15px rgba(0,0,0,0.1);
}

header::before {
    content: "";
    position: absolute;
    top: 0;
    left: 0;
    width: 100%;
    height: 5px;
    background: linear-gradient(90deg,
        #8E44AD 0%,
        #E74C3C 20%,
        #27AE60 40%,
        #F39C12 60%,
```

```
        #3498DB 80%,
        #16A085 100%);
}

h1 {
    font-family: 'Amiri', serif;
    font-size: 3.5rem;
    margin-bottom: 0.5rem;
    color: var(--gold);
    position: relative;
    display: inline-block;
    text-shadow: 1px 1px 3px rgba(0,0,0,0.3);
}

.subtitle {
    font-size: 1.3rem;
    opacity: 0.9;
    max-width: 800px;
    margin: 0 auto 1.5rem;
    font-family: 'Amiri', serif;
}

.genre-tabs {
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 1rem;
    margin-bottom: 3rem;
}

.genre-tab {
    padding: 0.8rem 1.5rem;
    border-radius: 30px;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.3s ease;
    color: white;
    box-shadow: 0 3px 10px rgba(0,0,0,0.1);
}

.genre-tab:hover {
    transform: translateY(-3px);
    box-shadow: 0 5px 15px rgba(0,0,0,0.2);
}

.genre-section {
    margin-bottom: 4rem;
```

```
    padding: 2rem;
    background: white;
    border-radius: 10px;
    box-shadow: 0 5px 15px rgba(0,0,0,0.05);
}

.genre-title {
    font-family: 'Amiri', serif;
    font-size: 2.5rem;
    margin-bottom: 1.5rem;
    padding-bottom: 0.5rem;
    border-bottom: 3px solid;
    display: inline-block;
}

.genre-description {
    max-width: 800px;
    margin-bottom: 2rem;
    font-size: 1.1rem;
    line-height: 1.8;
    color: #555;
}

.artists-container {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(350px, 1fr));
    gap: 2.5rem;
    margin-top: 2rem;
}

.artist-card {
    background: var(--white);
    border-radius: 12px;
    overflow: hidden;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.08);
    transition: all 0.3s ease;
    position: relative;
    border-top: 5px solid;
}

.artist-card:hover {
    transform: translateY(-8px);
    box-shadow: 0 15px 40px rgba(0, 0, 0, 0.15);
}

.artist-header {
    padding: 1.8rem;
```

```
    position: relative;
    color: white;
}

.artist-header::after {
    content: "";
    position: absolute;
    bottom: 0;
    left: 0;
    width: 100%;
    height: 4px;
    background: rgba(255,255,255,0.3);
}

.artist-name {
    font-family: 'Playfair Display', serif;
    font-size: 1.8rem;
    margin-bottom: 0.5rem;
    display: flex;
    align-items: center;
}

.artist-name i {
    margin-right: 10px;
    font-size: 1.5rem;
}

.artist-city {
    display: inline-block;
    background: rgba(255, 255, 255, 0.2);
    padding: 0.4rem 1rem;
    border-radius: 20px;
    font-size: 0.9rem;
    margin-top: 0.5rem;
}

.artist-body {
    padding: 2rem;
}

.info-item {
    margin-bottom: 1.2rem;
}

.info-label {
    font-weight: 600;
    display: block;
```

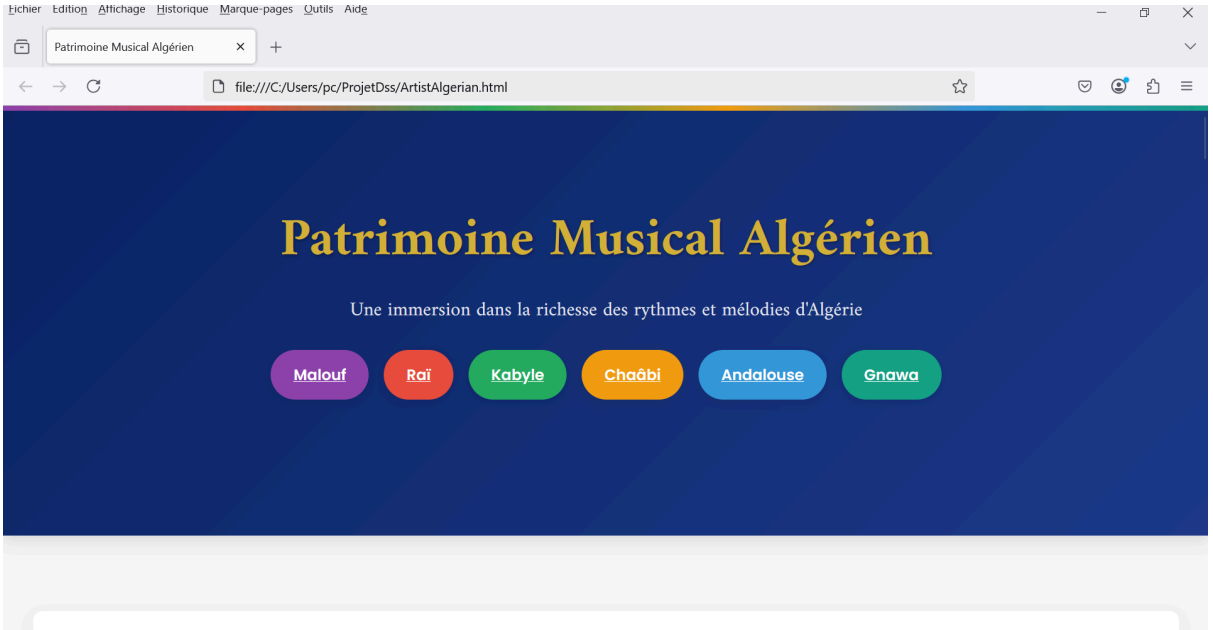
```
    margin-bottom: 0.3rem;
    font-size: 0.9rem;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}

.info-value {
    color: #555;
    font-size: 1.1rem;
}

.info-value.site {
    color: var(--dark-blue);
    text-decoration: none;
    font-weight: 500;
    transition: color 0.3s ease;
    word-break: break-all;
}

.info-value.site:hover {
    color: var(--light-blue);
    text-decoration: underline;
}

.album-section {
    margin-top: 2rem;
    padding-top: 1.5rem;
    border-top: 1px solid #eee;
}

.album-title {
    font-family: 'Playfair Display', serif;
    font-size: 1.4rem;
    margin-bottom: 1.2rem;
    display: flex;
    align-items: center;
}

.album-title::before {
    content: "🎵";
    margin-right: 0.8rem;
    font-size: 1.2rem;
}

.song-list {
    list-style: none;
}
```

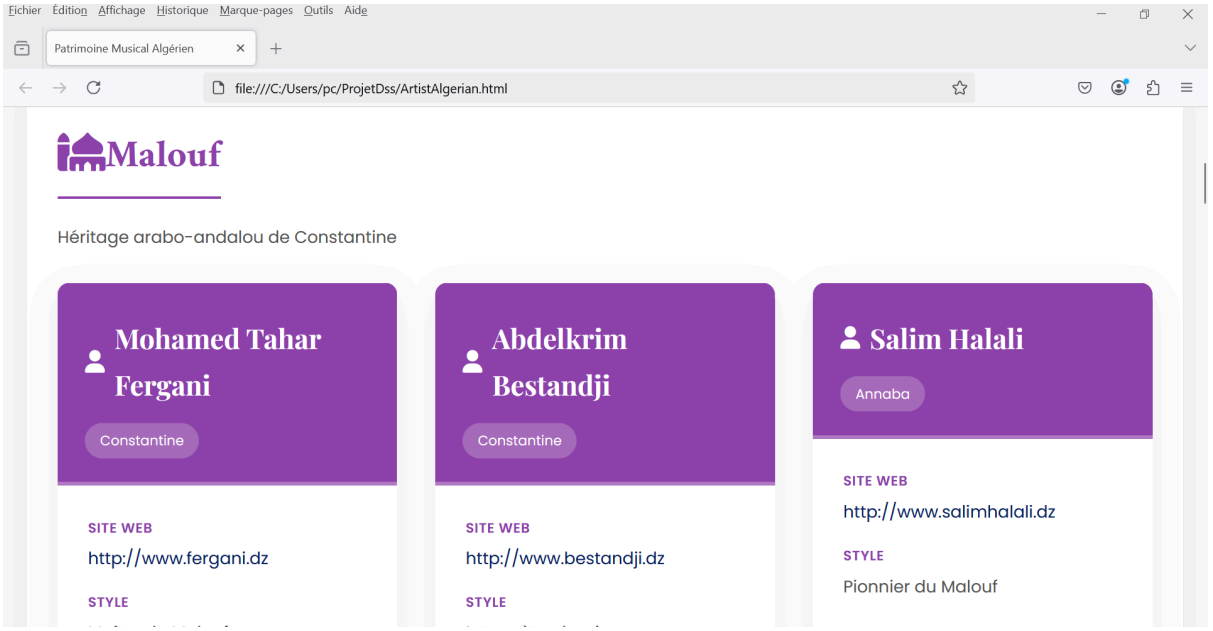
```
        .song-item {
            padding: 0.7rem 0;
            border-bottom: 1px dashed #ddd;
            display: flex;
            align-items: center;
            transition: all 0.2s ease;
        }

        .song-item:hover {
            background-color: rgba(0,0,0,0.02);
            padding-left: 5px;
        }

        .song-item::before {
            content: "♫";
            margin-right: 0.8rem;
            font-size: 0.9rem;
        }

        .song-item:last-child {
            border-bottom: none;
        }

        footer {
            text-align: center;
            margin-top: 4rem;
            padding: 2rem;
            color: #666;
            font-size: 0.9rem;
            border-top: 1px solid #eee;
            background: white;
        }

        @media (max-width: 768px) {
            .artists-container {
                grid-template-columns: 1fr;
            }

            h1 {
                font-size: 2.4rem;
            }
        }
    </style>
</head>
<body>
    <header>
```

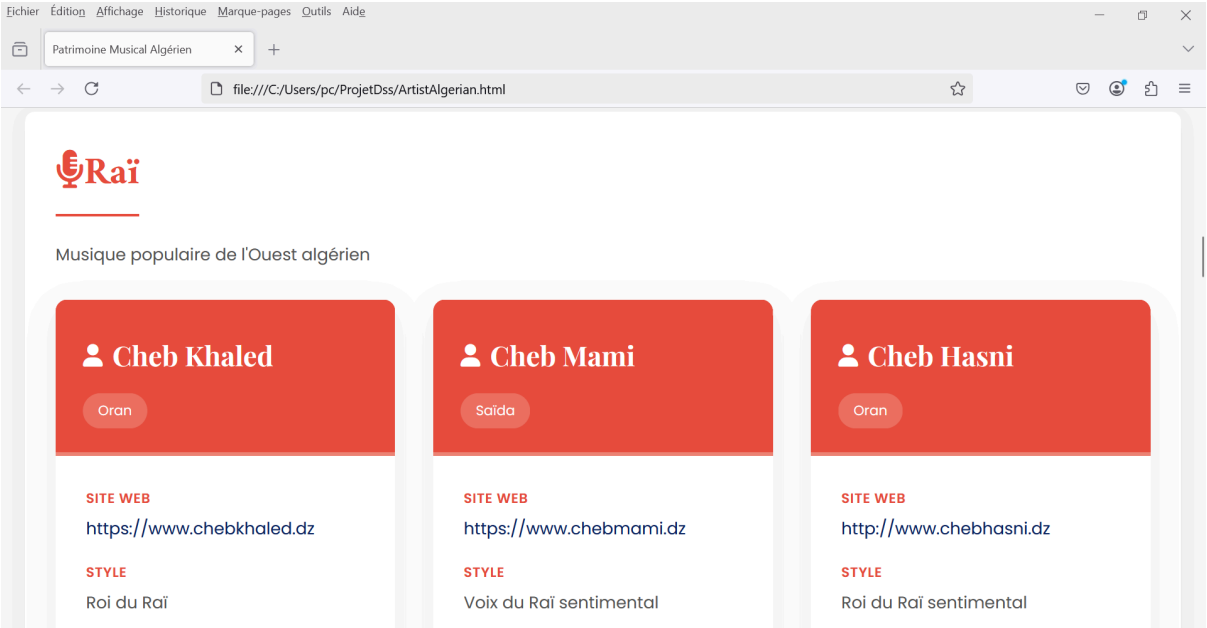
```
            <div class="container">
              <h1>Patrimoine Musical Algérien</h1>
              <p class="subtitle">Une immersion dans la richesse des rythmes et
mélodies d'Algérie</p>

              <div class="genre-tabs">
                <xsl:for-each select="artists/genre">
                  <a href="#{translate(@name, ' ', '')}" class="genre-tab"
style="background-color: {@color}">
                    <xsl:value-of select="@name"/>
                  </a>
                </xsl:for-each>
              </div>
            </div>
          </header>

          <div class="container">
            <xsl:apply-templates select="artists/genre"/>

            <footer>
              <div class="container">
                <p>© Culture Algérienne | Tous droits réservés</p>
              </div>
            </footer>
          </div>
        </body>
      </html>
    </xsl:template>

    <xsl:template match="genre">
      <div class="genre-section" id="{translate(@name, ' ', '')}">
        <h2 class="genre-title" style="color: {@color}; border-color: {@color}">
          <i class="fas fa-{@icon}"></i> <xsl:value-of select="@name"/>
        </h2>
        <p class="genre-description"><xsl:value-of select="@description"/></p>

        <div class="artists-container">
          <xsl:apply-templates select="artist">
            <xsl:with-param name="genre-color" select="@color"/>
          </xsl:apply-templates>
        </div>
      </div>
    </xsl:template>

    <xsl:template match="artist">
      <xsl:param name="genre-color"/>
```

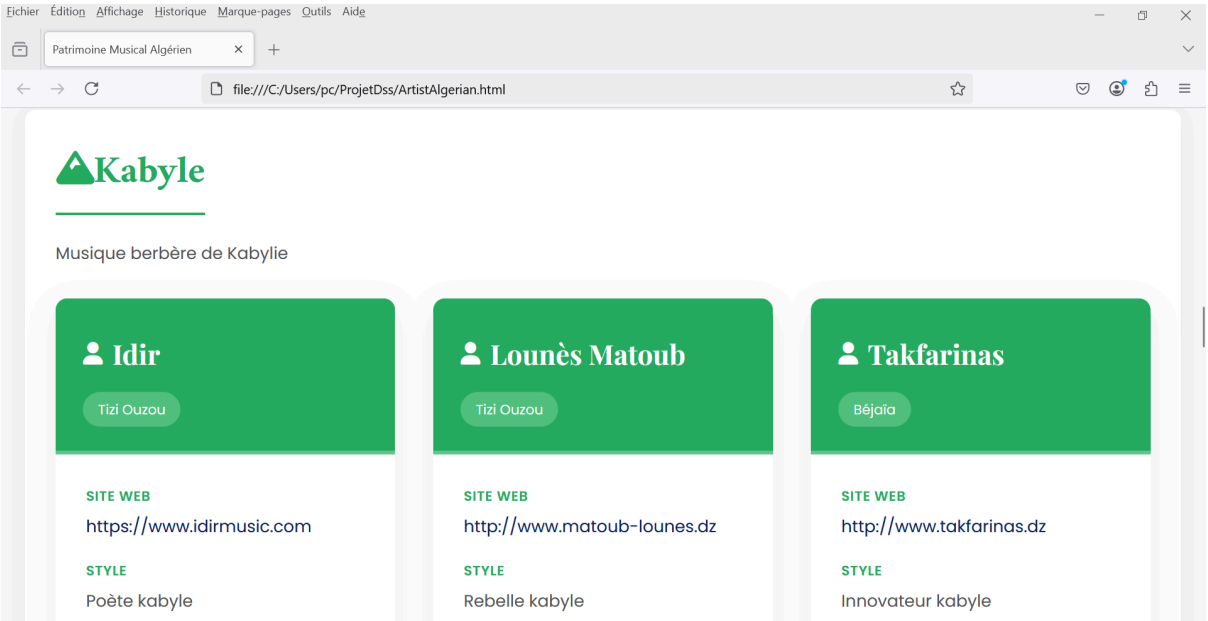
```
      <div class="artist-card" style="border-color: {$genre-color}">
         <div class="artist-header" style="background-color: {$genre-color}">
            <h2 class="artist-name"><i class="fas fa-user"></i> <xsl:value-of
select="name"/></h2>
            <span class="artist-city"><xsl:value-of select="city"/></span>
         </div>

         <div class="artist-body">
            <div class="info-item">
               <span class="info-label" style="color: {$genre-color}">Site Web</span>
               <a href="{site}" class="info-value site" target="_blank"><xsl:value-of
select="site"/></a>
            </div>

            <div class="info-item">
               <span class="info-label" style="color: {$genre-color}">Style</span>
               <span class="info-value"><xsl:value-of select="production"/></span>
            </div>

            <xsl:for-each select="albums/album">
               <div class="album-section">
                  <h3 class="album-title" style="color: {$genre-color}"><xsl:value-of
select="name"/></h3>
                  <ul class="song-list">
                     <xsl:for-each select="songs/song">
                        <li class="song-item" style="color: {$genre-color}"><xsl:value-of
select="."/></li>
                     </xsl:for-each>
                  </ul>
               </div>
            </xsl:for-each>
         </div>
      </div>
   </xsl:template>
</xsl:stylesheet>
```

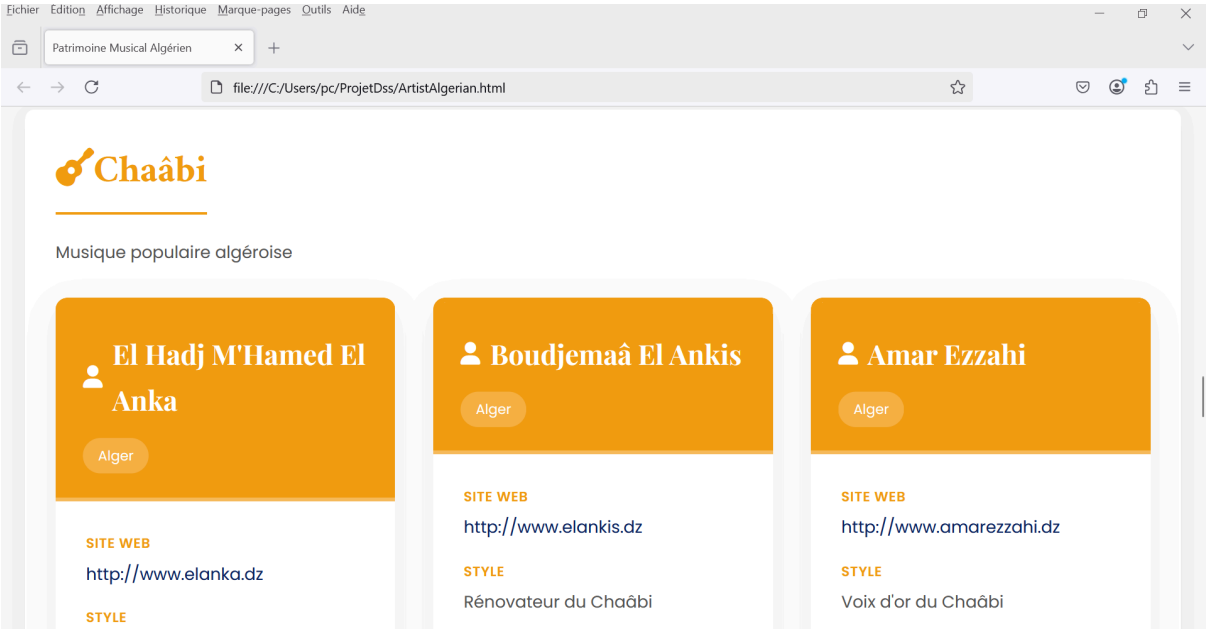
## 3.4 Expected Output

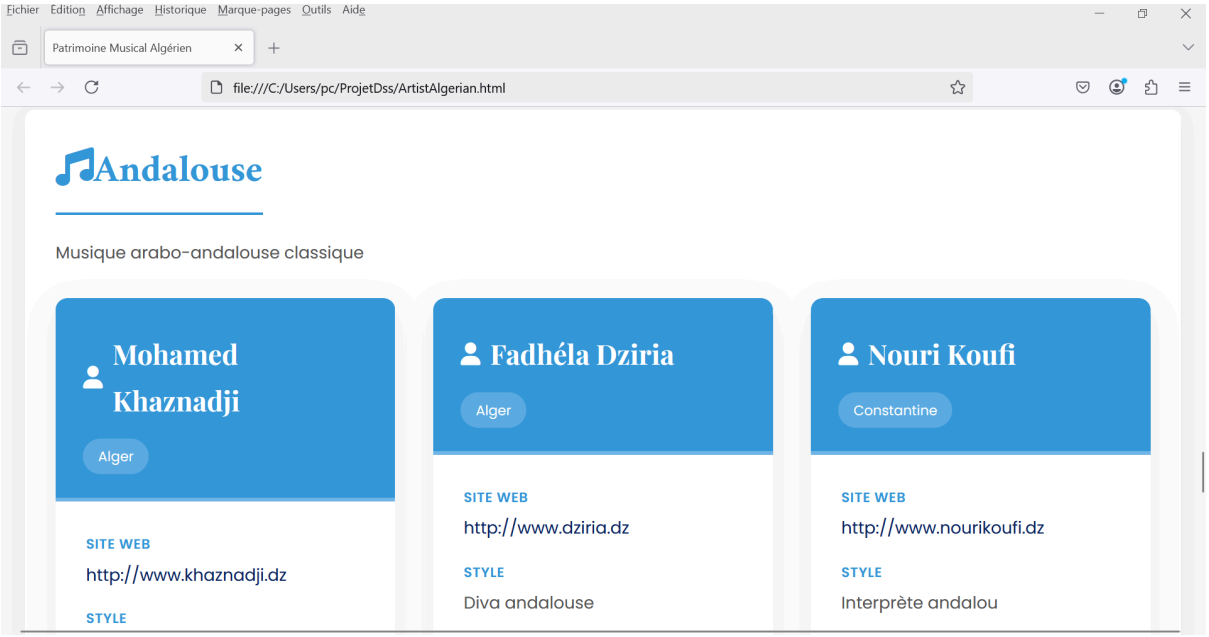


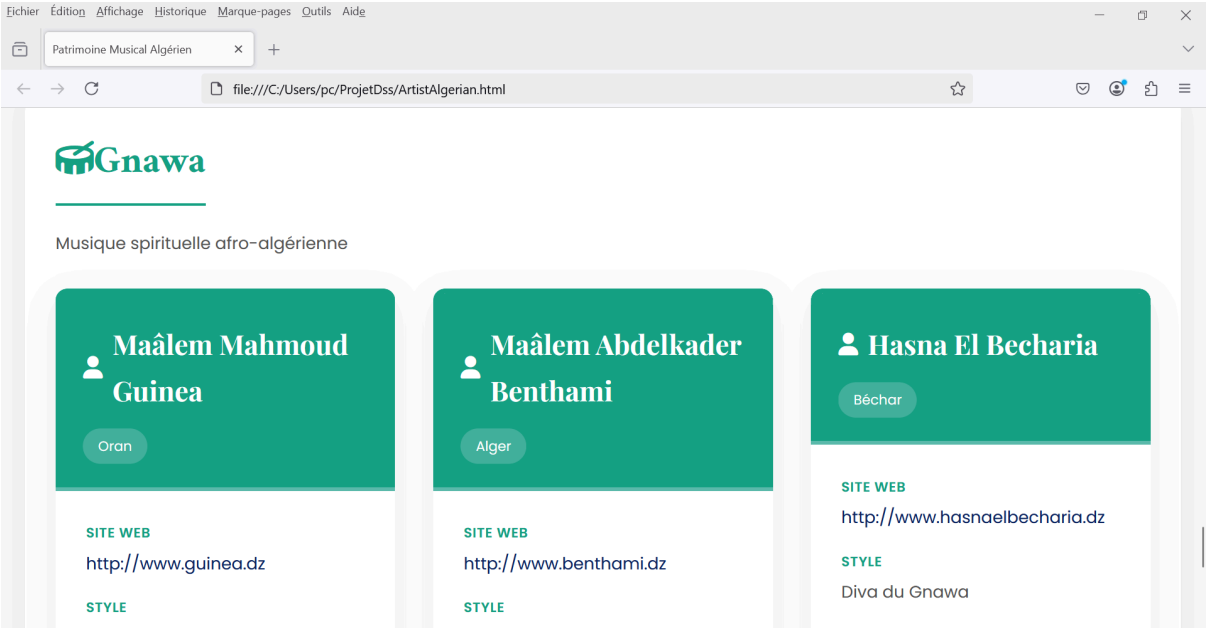



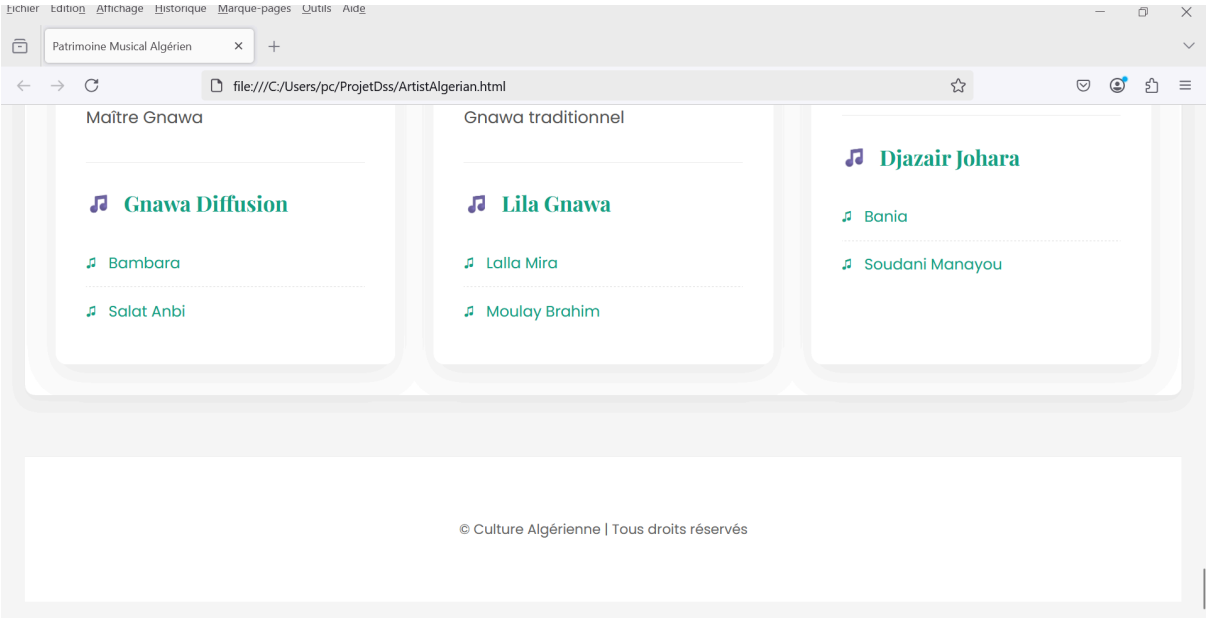

Fichier Édition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
file:///C:/Users/pc/ProjetDss/ArtistAlgerian.html
Raï
Musique populaire de l'Ouest algérien
Cheb Khaled
Oran
SITE WEB
https://www.chebkhaled.dz
STYLE
Roi du Raï
Cheb Mami
Saïda
SITE WEB
https://www.chebmami.dz
STYLE
Voix du Raï sentimental
Cheb Hasni
Oran
SITE WEB
http://www.chebhasni.dz
STYLE
Roi du Raï sentimental


Fichier Édition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
file:///C:/Users/pc/ProjetDss/ArtistAlgerian.html
Kabyle
Musique berbère de Kabylie
Idir
Tizi Ouzou
SITE WEB
https://www.idirmusic.com
STYLE
Poète kabyle
Lounès Matoub
Tizi Ouzou
SITE WEB
http://www.matoub-lounes.dz
STYLE
Rebelle kabyle
Takfarinas
Béjaïa
SITE WEB
http://www.takfarinas.dz
STYLE
Innovateur kabyle


Fichier Edition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
file:///C:/Users/pc/ProjetDss/ArtistAlgerian.html
Chaâbi
Musique populaire algéroise
El Hadj M'Hamed El Anka
Alger
SITE WEB
http://www.elanka.dz
STYLE
Boudjemaâ El Ankis
Alger
SITE WEB
http://www.elankis.dz
STYLE
Rénovateur du Chaâbi
Amar Ezzahi
Alger
SITE WEB
http://www.amarezzahi.dz
STYLE
Voix d'or du Chaâbi


Fichier Edition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
file:///C:/Users/pc/ProjetDss/ArtistAlgerian.html
Andalouse
Musique arabo-andalouse classique
Mohamed Khaznadji
Alger
SITE WEB
http://www.khaznadji.dz
STYLE
Fadhéla Dziria
Alger
SITE WEB
http://www.dziria.dz
STYLE
Diva andalouse
Nouri Koufi
Constantine
SITE WEB
http://www.nourikoufi.dz
STYLE
Interprète andalou




# 4. Part 2 - Python Application

## 4.1 Architecture

/ProjetDs/

│── app.py # Main application

```
│── ArtistAlgerian.xml    # Artist data
│── ArtistAlgerian.xsl    # XSLT transformation
└── templates/            # HTML templates
    ├── base.html         # Base template
    ├── results.html      # Search results
```

## 4.2 Implemented Features

- ✓ Artist name search
- ✓ Detail display (city, biography)
- ✓ Album and track listings
- ✓ Integrated XSLT transformation

## 4.3 Flask Code Excerpt

```
from flask import Flask, render_template, request, redirect, url_for
from lxml import etree
import os

app = Flask(__name__)

# Configuration
app.config['XML_FILE'] = 'ArtistAlgerian.xml'
app.config['XSL_FILE'] = 'ArtistAlgerian.xsl'

def parse_xml():
    try:
        if not os.path.exists(app.config['XML_FILE']):
            raise FileNotFoundError(f"Fichier XML {app.config['XML_FILE']} introuvable")
        return etree.parse(app.config['XML_FILE'])
    except Exception as e:
        app.logger.error(f"Erreur lors du parsing XML: {str(e)}")
        return None

def search_artist(name):
    try:
        tree = parse_xml()
        if tree is None:
            return None
```

```
        artists = []
        for artist in tree.xpath(f"//artist[contains(translate(name,
'ABCDEFGHIJKLMNOPQRSTUVWXYZ', 'abcdefghijklmnopqrstuvwxyz'),
translate('{name}', 'ABCDEFGHIJKLMNOPQRSTUVWXYZ',
'abcdefghijklmnopqrstuvwxyz'))]"):
            genre = artist.xpath("./ancestor::genre")[0]
            artist_data = {
                'name': artist.find('name').text,
                'city': artist.find('city').text,
                'site': artist.find('site').text if artist.find('site') is not None else '',
                'production': artist.find('production').text,
                'genre': {
                    'name': genre.get('name'),
                    'color': genre.get('color'),
                    'icon': genre.get('icon'),
                    'description': genre.get('description')
                },
                'albums': [],
                'biography': generate_biography(
                    artist.find('name').text,
                    artist.find('city').text,
                    artist.find('production').text
                )
            }

            for album in artist.xpath('./albums/album'):
                album_data = {
                    'name': album.find('name').text,
                    'songs': [song.text for song in album.xpath('./songs/song')]
                }
                artist_data['albums'].append(album_data)

            artists.append(artist_data)

        return artists
    except Exception as e:
        app.logger.error(f"Erreur lors de la recherche: {str(e)}")
        return None

def generate_biography(name, city, style):
    return f"{name} est un artiste renommé originaire de {city}, spécialisé dans
{style}. Avec une carrière prolifique, il a marqué la scène musicale algérienne par
son talent unique et son style distinctif."

@app.route('/')
def index():
    try:
```

```
        if os.path.exists(app.config['XML_FILE']) and
os.path.exists(app.config['XSL_FILE']):
            xml = etree.parse(app.config['XML_FILE'])
            xslt = etree.parse(app.config['XSL_FILE'])
            transform = etree.XSLT(xslt)
            html_content = str(transform(xml))
            return render_template('index.html', content=html_content)
        else:
            return render_template('error.html', message="Fichiers XML ou XSL
manquants"), 404
    except Exception as e:
        app.logger.error(f"Erreur dans la route index: {str(e)}")
        return render_template('error.html', message="Erreur de traitement des
données"), 500

@app.route('/search', methods=['GET', 'POST'])
def search():
    try:
        if request.method == 'POST':
            query = request.form.get('query', '').strip()
        else:
            query = request.args.get('query', '').strip()

        if not query:
            return redirect(url_for('index'))

        artists = search_artist(query)
        if artists is None:
            return render_template('error.html', message="Erreur lors de la recherche"),
500

        return render_template('search_results.html',
                               artists=artists,
                               query=query)
    except Exception as e:
        app.logger.error(f"Erreur dans la route search: {str(e)}")
        return render_template('error.html', message="Erreur lors de la recherche"),
500

@app.route('/artist/<name>')
def artist_detail(name):
    try:
        artists = search_artist(name)
        if not artists:
            return render_template('error.html', message="Artiste non trouvé"), 404

        return render_template('artist_detail.html', artist=artists[0])
```

```
    except Exception as e:
        app.logger.error(f"Erreur dans la route artist_detail: {str(e)}")
        return render_template('error.html', message="Erreur lors de l'affichage de
l'artiste"), 500

@app.errorhandler(404)
def page_not_found(e):
    return render_template('error.html', message="Page non trouvée"), 404

@app.errorhandler(500)
def internal_server_error(e):
    return render_template('error.html', message="Erreur interne du serveur"), 500

if __name__ == '__main__':
    app.run(debug=True)
```

# 5. User Guide

## 5.1 Installation

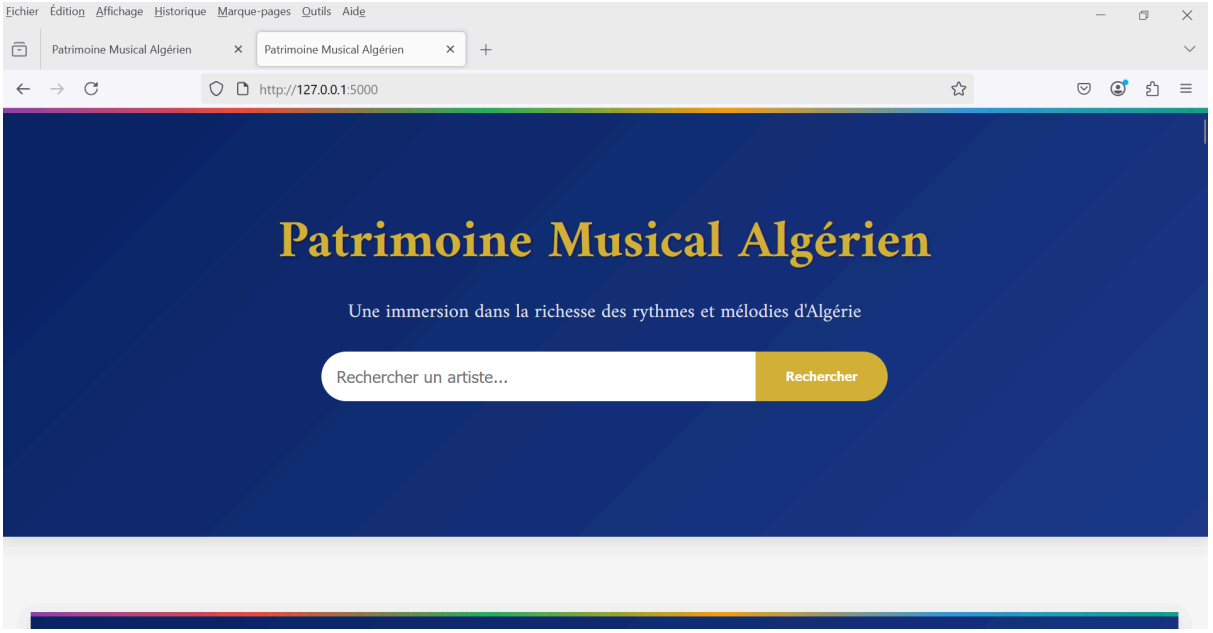
**pip install flask lxml**

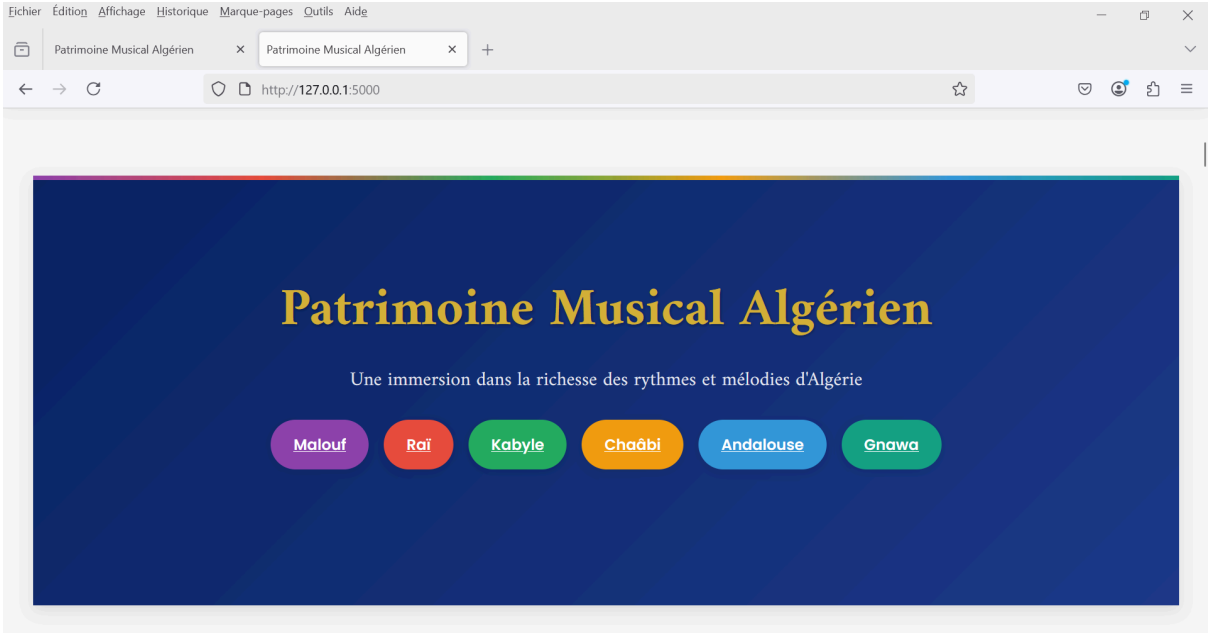
## 5.2 Execution

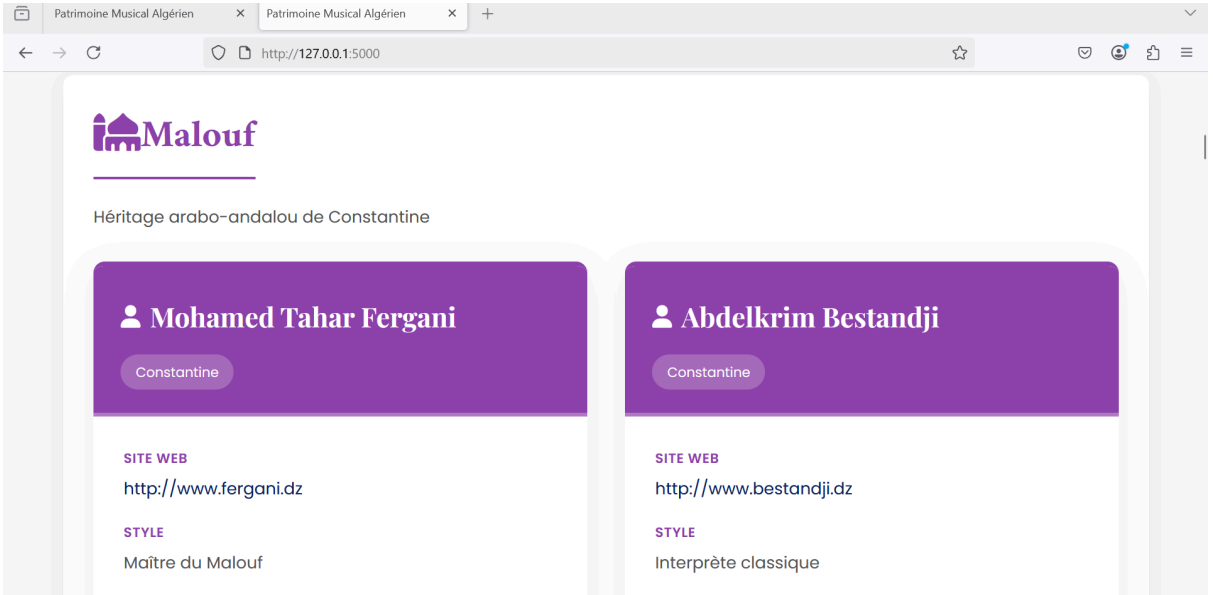
**python app.py**

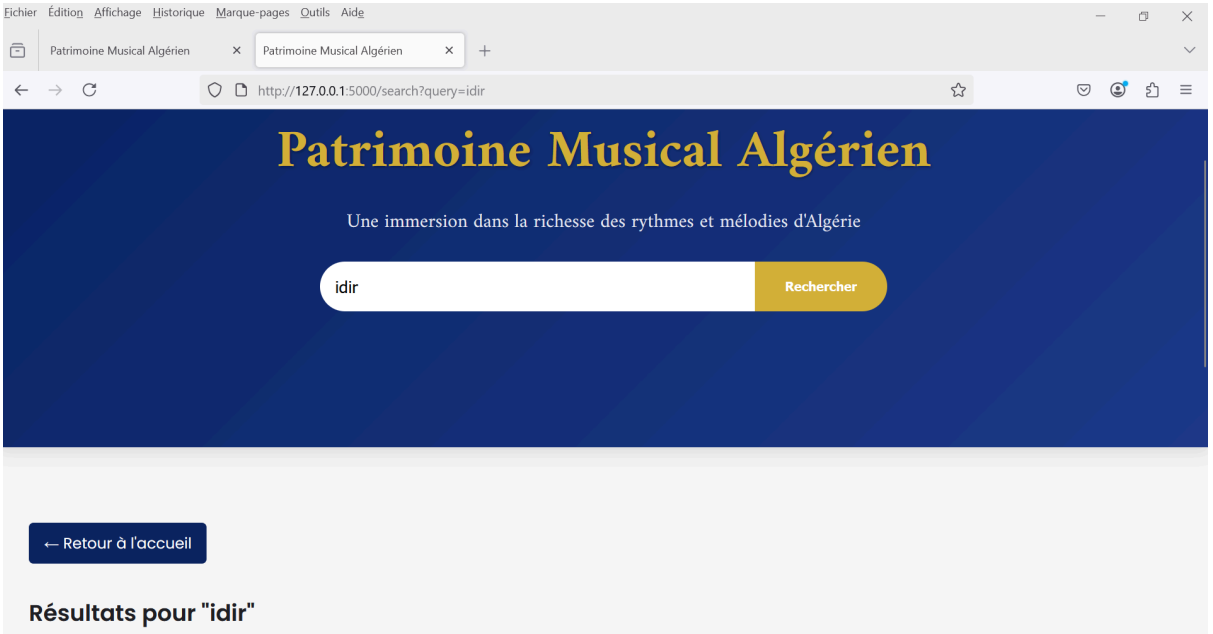
## 5.2 Expected Output

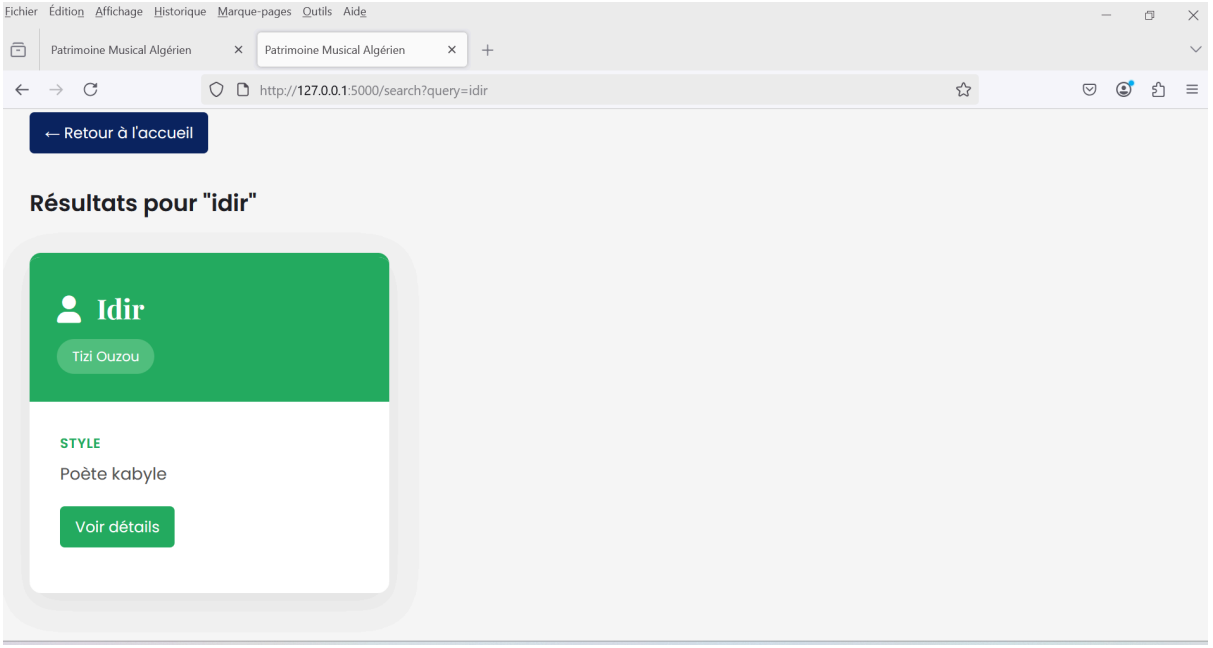
**Running on http://127.0.0.1:5000**

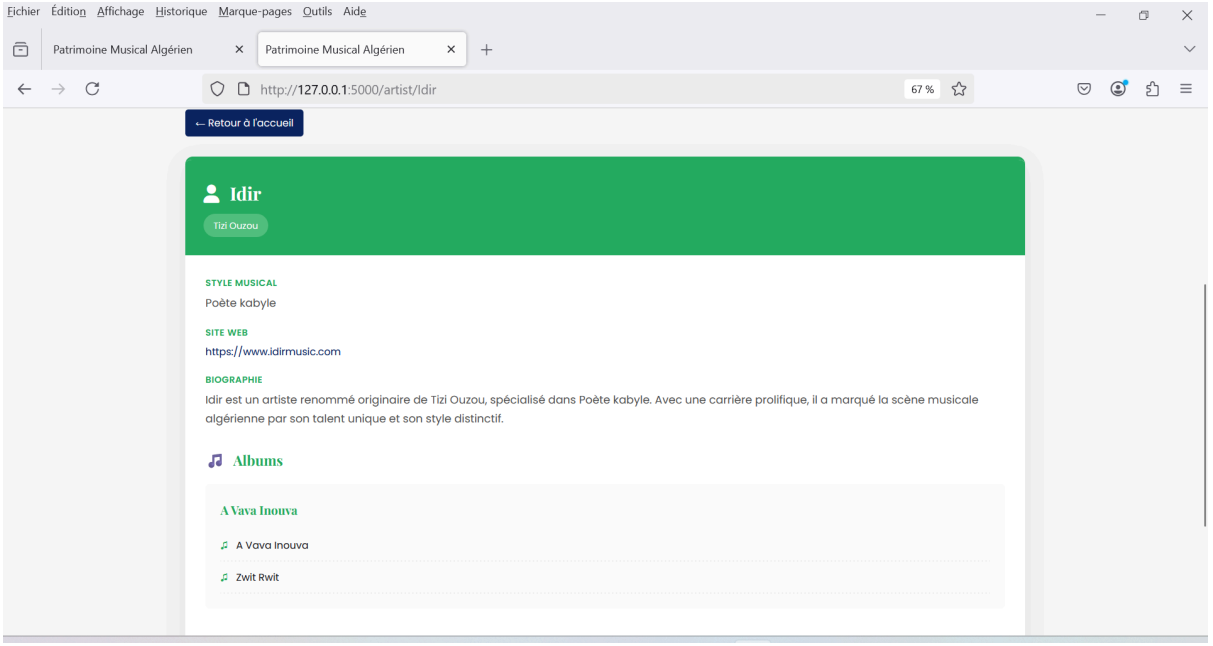

Fichier Édition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
Patrimoine Musical Algérien
http://127.0.0.1:5000
Patrimoine Musical Algérien
Une immersion dans la richesse des rythmes et mélodies d'Algérie
Rechercher un artiste...
Rechercher


Fichier Édition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
Patrimoine Musical Algérien
http://127.0.0.1:5000
Patrimoine Musical Algérien
Une immersion dans la richesse des rythmes et mélodies d'Algérie
Malouf
Raï
Kabyle
Chaâbi
Andalouse
Gnawa


Patrimoine Musical Algérien
Patrimoine Musical Algérien
http://127.0.0.1:5000
Malouf
Héritage arabo-andalou de Constantine
Mohamed Tahar Fergani
Constantine
SITE WEB
http://www.fergani.dz
STYLE
Maître du Malouf
Abdelkrim Bestandji
Constantine
SITE WEB
http://www.bestandji.dz
STYLE
Interprète classique


Fichier Édition Affichage Historique Marque-pages Outils Aide
Patrimoine Musical Algérien
Patrimoine Musical Algérien
http://127.0.0.1:5000/search?query=idir
Patrimoine Musical Algérien
Une immersion dans la richesse des rythmes et mélodies d'Algérie
idir
Rechercher
← Retour à l'accueil
Résultats pour "idir"




# II.Conclusion

By implementing this dual-component system, we establish a foundation for preserving Algeria's musical legacy digitally. The project demonstrates how traditional cultural assets can leverage modern web technologies for enhanced accessibility and engagement. Future expansions could include audio samples and interactive timelines.